

# Mazen Emam

Software Engineer · Frontend Specialist

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## PROFILE

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Software Engineer specializing in frontend development with 4 years of experience building scalable web applications that drive business results. I help companies enhance user experiences and operational efficiency through data-driven solutions, having contributed to platforms serving manufacturing, analytics, and recruitment sectors across MENA and Europe.

## WORK EXPERIENCE

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[Siren Analytics](#) · Beirut, Lebanon

Jun 2025 – Jan 2026

*Software Engineer*

**Stack:** React.js · TypeScript · Material UI · React Query · Redux Toolkit · WebSocket · SockJS/STOMP · React Hook Form · Yup

- Led development of an AI chat module enabling users to query 300+ news sources conversationally, replacing manual content navigation across bulk-ingested datasets.
- Implemented role-based access control across a multi-tenant architecture, scoping feature and content visibility per tenant and enforcing user-level permissions platform-wide.
- Engineered a multi-language translation system with React Query caching and bulk translation hooks, supporting the company's expansion into the European market.
- Pioneered a real-time WebSocket notification system using SockJS/STOMP with throttled error handling, enabling instant cross-user threat flagging that navigates recipients directly to flagged content.
- Optimized performance by integrating React Compiler and leading application-wide code splitting, reducing unnecessary re-renders and improving bundle efficiency.
- Secured authentication by refactoring JWT expiry logic and eliminating race conditions in token refresh, preventing unauthorized access.
- Eliminated a class of runtime errors by establishing a TypeScript enforcement pipeline with strict ESLint rules and a pre-push hook, catching type inconsistencies before code integration.

[Garment IO](#) · Giza, Egypt

May 2023 – Jun 2025

*Mid-Senior Frontend Engineer*

**Stack:** React.js · TypeScript · Material UI · Redux · Sentry · Formik · Yup · AWS · GitHub Actions

- Introduced TypeScript and strict ESLint enforcement across a 6-year-old codebase spanning 10+ modules and 40+ pages via an incremental migration strategy, establishing type safety without disrupting active development.
- Modernized the build pipeline by migrating from Create React App to Vite, reducing production build time by 50% and improving local development speed by 5x.
- Designed an i18n system with real-time language switching across 4 languages, enabling product deployment across MENA and European markets.
- Implemented Sentry performance monitoring capturing Web Vitals and API latency bottlenecks, identifying production issues across 30+ manufacturing facilities.
- Built a QR code scanning system for line managers, embedding machine IDs into scannable codes and eliminating manual lookup across 30+ facilities.
- Resolved performance bottlenecks and memory leaks by converting sequential API calls to parallel execution, removing unused data fetching, and enforcing pagination on unbounded endpoints.

*Frontend Engineer*

**Stack:** React.js · TypeScript · Material UI · MobX · React Hook Form · Yup

- Established a reusable component library with RTL support, creating a consistent design system that reduced new feature development time by 50%.
- Executed full application internationalization with dynamic RTL switching, enabling product launch in Arabic and English markets.
- Accelerated app startup speed by 35% through image optimization, lazy loading, and efficient API integrations.
- Developed a multi-step registration form using React Context, submitting mandatory fields per step, reducing backend load from incomplete submissions.

**Akwad** · Alexandria, Egypt

Oct 2021 – Jul 2022

*Frontend Engineer*

**Stack:** React.js · JavaScript · Bootstrap · Context API · React Hook Form

- Created a reusable form and grid component framework, standardizing development and reducing page build time by 90%.
- Delivered a complex Master-Detail system for asset and order management, enabling critical business workflows.
- Overhauled the core data grid infrastructure, unifying pagination and search logic to eliminate inconsistent behavior and reduce bugs.
- Replaced silent failures with a centralized error handling system featuring user-facing toast notifications and admin oversight logs.

## PROJECTS

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### Echo FM — Autonomous AI Satirical Radio Station

[GitHub](#) – [Live](#)

**Stack:** Python · GitHub Actions · Supabase · Webhooks · LLM · FFmpeg · Serverless Architecture

- Designed and shipped a fully autonomous, serverless media pipeline: live news ingestion → LLM script generation → TTS synthesis → LUFS-mastered audio → automated YouTube publishing, running on a 6-hour CRON schedule with zero manual intervention.
- Architected a resilient multi-service event-driven system across GitHub Actions, Supabase, and Netlify with webhook-driven deployment, retry logic, custom JSON repair for truncated LLM responses, and multi-tier engine fallback chains.
- Integrated multiple LLM APIs (Gemini, Groq/Llama-4) with a custom Smart Router and quality validator; built a 3-tier TTS cascade enforcing vocal consistency across 13-segment broadcasts via a Unified Master Engine pattern.
- Designed the station's visual identity and frontend dashboard, embedding the design system into both the web UI and the automated cover art pipeline.

### Job Radar — AI-Powered Job Dashboard

[GitHub](#) – [Live](#)

**Stack:** Next.js · TypeScript · Tailwind CSS · Supabase · Gemini API

- Built AI filtering UI (Regex + Culture Detector + Gemini) cutting job noise by ~85%, surfacing high-match Senior React/Next.js roles with transparent scoring.
- Architected modular Next.js frontend with centralized constants/types, standardizing scoring across pipelines and cutting feature dev time by ~40%.
- Designed seniority-aware routing dynamically adjusting gates per pipeline, boosting opportunities to surface international roles.
- Built a Market Intelligence dashboard with trend analysis, skill co-occurrence charts, and missing-skill insights using React and Supabase.
- Implemented production safeguards via CRON\_SECRET validation, secret isolation, and optimistic UI updates, securing workflows without sacrificing UX.

## TECHNOLOGIES

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**Proficient** React.js · TypeScript · JavaScript · Redux/Redux Toolkit · React Query/TanStack Query · Material-UI · Vite · HTML · CSS · SASS · Bootstrap · Git

**Familiar** Next.js · Node.js · Express · MongoDB · GraphQL · WebSocket/STOMP · MobX · Sentry · Jira · Jest · Vitest · React Testing Library · AWS · Docker · Webpack

## EDUCATION

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### Alexandria University

Oct 2013 – Jul 2017

*BSc. in Management Information Systems, Faculty of Commerce*

### Certificates

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#### Udacity

Jul 2022 – Aug 2022

[React Nanodegree](#)

#### freeCodeCamp

Jun 2022 – Jul 2022

[JavaScript Algorithms and Data Structures](#)

#### Information Technology Institute (ITI)

Sep 2020 – Dec 2020

[MEARN Track — 3 Month Diploma](#)

## LANGUAGES

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**Arabic** — Native | **English** — Fluent (C1)